PHARVARIS

Summary of Results

A Phase 3, Randomized, Double-Blind, Placebo-Controlled, Cross-Over Study Of Oral Deucrictibant Soft Capsule For On-Demand Treatment Of Attacks In Adolescents And Adults With Hereditary Angioedema

EU CT Number: 2023-507268-37-00

Sponsor

Pharvaris Netherlands B.V.
J.H. Oortweg 21
2333 CH Leiden
The Netherlands

Study ID: 2023-507268-37-00

Sponsor Name: Pharvaris Netherlands B.V.
Study Molecule: Deucrictibant soft capsule

A.CLINICAL TRIAL INFORMATION

1. TRIAL IDENTIFICATION

Title of the Clinical Trial	A Phase 3, Randomized, Double-blind, Placebo-controlled, Cross-over Study of Oral Deucrictibant Soft Capsule for On-Demand Treatment of Attacks in Adolescents and Adults with Hereditary Angioedema
EU Clinical Trial Number	2023-507268-37-00
Sponsor protocol number	PHA022121-C306 (RAPIDe-3)
Type of Results Summary	Final

2. ADDITIONAL STUDY IDENTIFIERS

NCT number	NCT07266805
ISRCTN number	NA
WHO universal trial number (UTN)	U1111-1295-6497

3. SPONSOR(S)

Sponsor organisation name	Pharvaris Netherlands B.V.
Sponsor organisation address	Pharvaris Netherlands B.V. J.H. Oortweg 21 2333 CH Leiden The Netherlands
Public contact	Pharvaris Netherlands B.V. – Pharvaris clinical Phone: +31712036410 Email: clinical@pharvaris.com
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4. PAEDIATRIC REGULATORY DETAILS

Is trial part of an agreed paediatric investigation plan (PIP)	Yes
Does article 45 of REGULATION (EC) No 1901/2006 apply to this trial?	No
Does article 46 of REGULATION (EC) No 1901/2006 apply to this trial?	No

5. RESULT ANALYSIS STAGE

Analysis stage	Final

Date of interim/final analysis	30-Oct-2025
Global end of trial reached?	Yes
Global end of trial date	17-Oct-2025
Was the trial ended prematurely?	No

6. GENERAL INFORMATION ABOUT THE TRIAL

Main objective of the trial:
The Primary objective of the trial was to evaluate the efficacy of deucrictibant as on-demand treatment compared with placebo on the onset of symptom relief during hereditary angioedema (HAE) attacks.

Protection of trial subjects:
The conduct of this study and the processing of any personal data collected from each participant (or from a participant's healthcare professional or other relevant third-party sources) by the Sponsor or it's designee, the site and the Investigator for use in the study will fully adhere to the requirements set out in applicable data protection and medical privacy laws or regulations, including, without limitation, the General Data Protection Regulation (GDPR) EU 2016/679. The Sponsor or its designee shall ensure that at all times it has an appropriate legal basis for processing personal data under applicable data protection law.

Actual start date of recruitment	04-Apr-2024
Independent data monitoring committee (IDMC) involvement?	Yes
Long term follow-up planned?	No

7. POPULATION OF TRIAL SUBJECTS

Subjects enrolled per country

Summary of Results

Country	Number of subjects enrolled
Total EEA only	53
Argentina	5
Australia	1
Austria	5
Brazil	8
Bulgaria	7
Canada	8
Czech Republic	3
France	1
Germany	8
Hong Kong	8
Hungary	2
Italy	16
Japan	5
Netherlands	2
Poland	5
Puerto Rico	3
Saudi Arabia	1
South Africa	3
South Korea	5
Spain	1

Sweden	3
Turkey	4
United Kingdom	3
United States	27
North America	38
Rest of the world	43
Worldwide total number of subjects	134

Subjects enrolled per age group

In utero	0
Preterm newborn - gestational age < 37 wk	0
Newborns (0-27 days)	0
Infants and toddlers (28 days-23 months)	0
Children (2-11 years)	0
Adolescents (12 -17 years)	10
Adults (18-64 years)	116
Adults (65-84 years)	8
85 years and over	0

Participants enrolled per gender

Gender of subjects	
Female	76

Male	58

B. SUBJECT DISPOSITION

1. RECRUITMENT

Recruitment details: This Phase 3, multicenter, randomized, double-blind, placebo-controlled, 2-period, 2-treatment cross-over study evaluated the efficacy and safety of orally administered deucrictibant immediate-release (IR) soft capsules compared with placebo for the on-demand treatment of attacks in participants with HAE.

Pre-Assignment Period: A total of 134 participants were enrolled in the study.

Eligibility

Key Inclusion Criteria:

The following inclusion criteria had to be met for a subject to be eligible for inclusion in the study:

1. Provision of written informed consent. If the participant is a minor (i.e., <18 years of age or as determined by local law), consent will be obtained from the participant's parent/legally designated representative/guardian and written assent will be obtained from the participant, per country regulations.
2. Male or female, aged ≥12 to ≤75 years at the time of providing written informed consent/assent.
3. Diagnosis of HAE - Type1/2 (HAE-C1INH), and Type 3(HAE-nC1INH).
4. History of at least 2 HAE attacks in the last 3 months before screening.
5. Experience with using standard-of-care treatment to effectively manage on-demand treatment for HAE attacks.
6. For adolescent participants aged ≥12 to <18 years: body weight ≥40 kg.
7. Either was on a stable dose and regimen of long-term prophylactic therapy indicated for HAE for at least 6 months before the screening visit and intend to remain on the same dose for the duration of the study or was receiving only on-demand treatment for HAE, then had followed the protocol specified washout period before the screening visit.

Key Exclusion Criteria:

1. Any female who is pregnant, plans to become pregnant, or is breastfeeding.
2. Any diagnosis of angioedema other than HAE.
3. Any clinically significant comorbidity or systemic dysfunction that would interfere with the participant's safety or ability to participate in the study.
4. Use of attenuated androgens for short-term prophylaxis within 2 weeks before screening.
5. Abnormal hepatic function (aspartate aminotransferase [AST] >2×upper limit of normal [ULN], alanine aminotransferase [ALT] >2×ULN, or total bilirubin >1.5×ULN, or Child-Pugh class B or C).
6. Abnormal renal function (eGFR <60 ml/min/1.73 m^2).
7. Has received prior on-demand HAE treatment with deucrictibant.

2. POST-ASSIGNMENT PERIODS

Period title	Overall Study
Is this the Baseline Period	Yes
Allocation method:	Randomized
Blinding used	Quadruple (Participant, Care Provider, Investigator, Outcomes Assessor).

ARMS AND INVESTIGATIONAL MEDICINAL PRODUCTS

Arms	Investigational Medicinal Products
Arm Title1: Crossover treatment sequence 1: Placebo - Deucrictibant IR 20 milligrams (mg) **Arm Description:** Participants were randomized in a cross	**Investigational medicinal product name:** Placebo and Deucrictibant **Pharmaceutical forms:** Oral Capsule

over manner to receive matching placebo for first HAE attack, followed by oral administration of deucrictibant IR 20 mg soft capsules for second HAE attack. **Arm Type**: Experimental	**Route of Administration:** Oral **Dosage and administration details:** Matching placebo followed by 20mg deucrictibant IR soft capsule orally
Arm Title 2: Crossover treatment sequence 2: Deucrictibant IR 20 mg - Placebo **Arm Description:** Participants were randomized in a cross over manner to receive 20 mg of deucrictibant IR soft capsules orally for first HAE attack, followed by matching placebo for second HAE attack. **Arm Type:** Experimental	**Investigational medicinal product name:** Deucrictibant and Placebo **Pharmaceutical forms:** Oral Capsule **Route of Administration:** Oral **Dosage and administration details:** 20 mg deucrictibant IR soft capsule followed by matching placebo orally

Are the Arms mutually exclusive? Yes

Subject Disposition

	Placebo - Deucrictibant IR 20 mg	Deucrictibant IR 20 mg - Placebo	Total
Started	68	66	134
Completed	41 (60.3%)	47 (71.2%)	88 (65.7%)
Not Completed	27 (39.7%)	19 (28.8%)	46 (34.3)
Death	0 (0%)	0 (0%)	0 (0%)

Summary of Results

	Placebo - Deucrictibant IR 20 mg	Deucrictibant IR 20 mg - Placebo	Total
Lost to Follow-up	2 (2.9%)	0 (0%)	2 (1.5%)
Physician Decision	0 (0%)	0 (0%)	0 (0%)
Consent Withdrawn by Subject	3 (4.4%)	3 (4.5%)	6 (4.5%)
Adverse Event Non-Fatal	0 (0%)	0 (0%)	0 (0%)
Adverse Event Serious Fatal	0 (0%)	0 (0%)	0 (0%)
Lack of Efficacy	0 (0%)	0 (0%)	0 (0%)
Pregnancy	0 (0%)	0 (0%)	0 (0%)
Protocol Deviation	2 (2.9%)	0 (0%)	2 (1.5%)
Transferred to Other arms/groups	0 (0%)	0 (0%)	0 (0%)
Other (Study terminated or suspended by sponsor)	20 (29.4%)	15 (22.7%)	35 (26.1%)
Other (Randomization error)	0	1 (1.5%)	1 (0.7%)

Baseline characteristics

Baseline Analysis Population Description: All Randomized Participants

Summary of Results

Reporting groups

Reporting group title	Reporting group description
Placebo - Deucrictibant IR 20 mg	Participants were randomized in a cross over manner to receive matching placebo for the first HAE attack, followed by oral administration of deucrictibant IR 20 mg soft capsules for the second HAE attack.
Deucrictibant IR 20 mg - Placebo	Participants were randomized in a cross over manner to receive deucrictibant IR 20 mg soft capsules orally for the first HAE attack, followed by matching placebo for the second HAE attack.

Demographics and baseline characteristics

		Placebo - Deucrictibant IR 20 mg	Deucrictibant IR 20 mg - Placebo	Total
Number of subjects		68	66	134
Age, Continuous	Number Analyzed	68	66	134
	Mean (Standard Deviation) Unit of measure: Years	38.9 (15.22)	39.1 (14.30)	39.0 (14.72)
Sex: Female, Male	Number Analyzed	68	66	134
	Female	39	37	76
	Male	29	29	58
Ethnicity (NIH/OMB)	Number Analyzed	68	66	134

Summary of Results

		Placebo - Deucrictibant IR 20 mg	Deucrictibant IR 20 mg - Placebo	Total
	Hispanic or Latino	6	11	17
	Not Hispanic or Latino	62	53	115
	Unknown or Not Reported	0	2	2
Race/Ethnicity, Customized	Number Analyzed	68	66	134
	Asian	7	12	19
	Black or African American	7	3	10
	White	50	43	93
	Other	3	4	7
	Not Reported	1	3	4
	American Indian or Alaska Native	0	1	1
	Native Hawaiian or Other Pacific Islander	0	0	0

D.END POINTS

Reporting groups:

Reporting group title	Reporting group description
Placebo	Participants were randomized in a cross over manner to receive matching placebo orally either for the first HAE attack or for the second HAE attack
Deucrictibant IR 20 mg	Participants were randomized in a cross over manner to receive 20 mg of deucrictibant IR soft capsules orally either for the first HAE attack or for the second HAE attack

1. **Primary Endpoint:**

End point Title	Time to onset of symptom relief assessed by Patient Global Impression of Change (PGI-C) assessment
End point Description	PGI-C is a 7-point response scale designed to evaluate the change in the attack symptoms from pre-treatment (the last symptom assessment before study drug administration) over time with a 7-point response scale. Participants were asked to choose the response that best describes currently at the time of assessment the overall change in their HAE attack symptoms since they took the study drug: Much better, better, a little better, same, a little worse, worse and much worse. Time to onset of symptom relief is calculated as the time from study drug administration for treating an attack (the time of the first dose if a second dose is administered) to the earliest time when PGI-C rating achieves “a little better” or higher response (“better”, or “much better”) and is sustained for 2 consecutive timepoints. The difference in time to the event of interest between deucrictibant and placebo was tested using Gehan’s generalized Wilcoxon test. The Kaplan-Meier estimated median and percentiles of time to onset of symptom relief by PGI-C assessment is presented. The intercurrent event (ICE) was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.

Time Frame	Up to 12 hours post-treatment

Analysis Population Description: Primary Efficacy Analysis Set (PEAS) comprised of all randomized participants who completed both attack periods, namely, who had two attacks treated with study drug (one in each attack period).

End point values		**Placebo**	**Deucrictibant IR 20 mg**
Number of subjects analysed		88	88
Units: Hours			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	NE (5.82, NE)	1.28 (1.05, 1.52)

NE= Not Estimable indicates that median was not estimable as less than 50% achieved the endpoint within 12-hour analysis window

Analysis 1

Statistical Analysis Overview	Comparison Group Selection	Placebo, Deucrictibant 20 mg
	Comments	Not Applicable
	Type of Statistical Test	Other
	Comments	Not Applicable

Statistical Test of Hypothesis	P-Value	<0.0001
	Comments	Not Applicable
	Method	Gehan's generalized Wilcoxon test
	Comments	Not Applicable

1. **Secondary Endpoint:**

Endpoint Title	Percentage of participants with study drug-treated attacks who achieved PGI-C rating of at least a little better.
Endpoint Description	PGI-C is a 7-point response scale designed to evaluate the change in the attack symptoms from pre-treatment (the last symptom assessment before study drug administration) over time with a 7-point response scale. Participants were asked to choose the response that best describes currently at the time of assessment the overall change in their HAE attack symptoms since they took the study drug: Much better, better, a little better, same, a little worse, worse and much worse. Onset of symptom relief is considered achieved at 4h if for an attack, the last or the second-to-last available value of PGI-C prior to 4h post-treatment has achieved "A little better" or higher response ("Better", or "Much better") and is sustained for two consecutive timepoints. Attacks with no PGI-C recorded within 4h post-treatment is excluded from the analysis, unless there was an ICE for that attack, in which case the attack is considered as not achieving PGI-C "A little better" at 4h. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	At 4 hours post treatment

Analysis Population Description: PEAS.

Summary of Results

End point values		Placebo	Deucrictibant IR 20 mg
Number of subjects analysed		88	88
Units: Percentage of participants			
Measure Type	**Precision/ Dispersion type**		
Number	Not Applicable	27.6	83.1

2. Secondary Endpoint

Endpoint Title	Time to substantial symptom relief by PGI-C assessment.
Endpoint Description	PGI-C is a 7-point response scale designed to evaluate the change in the attack symptoms from pre-treatment (the last symptom assessment before study drug administration) over time with a 7-point response scale. Participants were asked to choose the response that best describes currently at the time of assessment the overall change in their HAE attack symptoms since they took the study drug: Much better, better, a little better, same, a little worse, worse and much worse. Time to substantial symptom relief is calculated as the time from study drug administration for treating an attack (the time of the first dose if a second dose is administered) to the earliest time when PGI-C rating achieves “Better” or higher response (“Much better”) and is sustained for 2 consecutive timepoints. The Kaplan-Meier estimated median and percentiles of time to onset of substantial symptom relief by PGI-C assessment is presented. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 12 hours post-treatment

Analysis Population Description: PEAS.

End point values		Placebo	Deucrictibant IR 20 mg
Number of subjects analysed		88	88
Units: Hours			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	NE (10.30, NE)	2.85 (2.08, 3.35)

NE= Not Estimable indicates that median was not estimable as less than 50% achieved the endpoint within 12-hour analysis window

3. Secondary Endpoint

Endpoint Title	Time to substantial symptom relief by Patient Global Impression of Severity (PGI-S) assessment.
Endpoint Description	PGI-S is designed to evaluate the severity of attack symptoms on a 5-point response scale. Participants were asked to choose the response that best describes the severity of their HAE attack symptoms currently at the time of assessment: None, mild, moderate, severe and very severe. Substantial symptom relief was achieved when there was a ≥1 point reduction in PGI-S from pre-treatment for 2 consecutive timepoints within 12 hours post-treatment. The Kaplan-Meier estimated median and percentiles of time to onset of substantial symptom relief by PGI-S assessment is presented. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 12 hours post-treatment

Analysis Population Description: PEAS.

End point values		Placebo	Deucrictibant IR 20 mg
Number of subjects analysed		88	88
Units: Hours			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	NE (12.34, NE)	2.41 (1.70, 2.98)

NE= Not Estimable indicates that median was not estimable as less than 50% achieved the endpoint within 12-hour analysis window

4. **Secondary Endpoint**

Endpoint Title	Time to complete symptom resolution by PGI-S assessment.
Endpoint Description	PGI-S is designed to evaluate the severity of attack symptoms on a 5-point response scale. Participants were asked to choose the response that best describes the severity of their HAE attack symptoms currently at the time of assessment: None, mild, moderate, severe and very severe. Complete symptom resolution was achieved when PGI-S rating of "none" within 48 hours post-treatment. The Kaplan-Meier estimated median and percentiles of time to complete symptom resolution by PGI-S assessment is presented. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 48 hours post-treatment

Analysis Population Description: PEAS.

End point values		Placebo	Deucrictibant IR 20 mg
Number of subjects analysed		88	88
Units: Hours			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	NE (NE, NE)	11.95 (8.61, 21.79)

NE= Not Estimable indicates that median was not estimable as less than 50% achieved the endpoint within 48-hours analysis window

5. Secondary Endpoint

Endpoint Title	Time to end of progression (EoP) in attack symptoms within 12 hours by PGI-C assessment
Endpoint Description	PGI-C is a 7-point response scale designed to evaluate the change in the attack symptoms from pre-treatment (the last symptom assessment before study drug administration) over time with a 7-point response scale. Participants were asked to choose the response that best describes currently at the time of assessment the overall change in their HAE attack symptoms since they took the study drug: Much better, better, a little better, same, a little worse, worse and much worse. Time to EoP in attack symptoms is calculated as the time from study drug administration for treating an attack (the time of the first dose if a second dose is administered) to the earliest post-treatment timepoint after which all subsequent PGI-C ratings are stable or improved within 12 hours post-treatment. The Kaplan-Meier estimated median and percentiles of time to EoP in attack symptoms by PGI-S assessment is presented. The ICE was the intake of prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.

Time Frame	Up to 12 hours post-treatment

Analysis Population Description: PEAS.

End point values		**Placebo**	**Deucrictibant IR 20 mg**
Number of subjects analysed		88	88
Units: Minutes			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	228.67 (61.82, 599.6)	17.47 (16.27, 19.17)

6. Secondary Endpoint

Endpoint Title	Percentage of participants with study drug-treated attacks requiring rescue medication
Endpoint Description	This binary endpoint ("Yes" or "No") was assigned a value of "Yes" for a study drug–treated attack if rescue medication was taken within 24 hours after study drug administration (defined as the time of the first dose administration if a second dose was administered). To ensure participant safety in the placebo-controlled study, rescue medication and global or locally available standard of care were allowed. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 24 hours post-treatment

Analysis Population Description: PEAS.

End point values		Placebo	Deucrictibant IR 20 mg
Number of subjects analysed		88	88
Units: Percentage of participants			
Measure Type	**Precision/ Dispersion type**		
Number	Not Applicable	44.3	9.1

7. Secondary Endpoint

Endpoint Title	Percentage of participants achieving symptom resolution with one dose of study drug after attacks by PGI-S assessment
Endpoint Description	PGI-S is designed to evaluate the severity of attack symptoms on a 5-point response scale. Participants were asked to choose the response that best describes the severity of their HAE attack symptoms currently at the time of assessment: None, mild, moderate, severe and very severe. Proportion of attacks achieving symptom resolution is defined as achieving PGI-S rating of "none" with one dose of study drug at 24 hours post-treatment. This binary endpoint ("Yes" or "No") was assigned a value of "Yes" if, for a study drug–treated attack, the last available PGI-S rating prior to 24 hours had achieved "None" and no second dose of the study drug had been taken prior to 24 hours. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 24 hours post-treatment

Analysis Population Description: PEAS.

Summary of Results

End point values		Placebo	Deucrictibant IR 20 mg
Number of subjects analysed		88	88
Units: Percentage of participants			
Measure Type	**Precision/ Dispersion type**		
Number	Not Applicable	14.9	60.2

8. Secondary endpoint

Endpoint Title	Time to substantial symptom relief by Angioedema Symptom Rating Scale (AMRA) assessment
Endpoint Description	The AMRA-3 is a composite score (average) of the 3 major HAE symptoms: skin swelling, skin pain, and abdominal pain. The AMRA-3 scale allowed scoring on a scale between 0 (No swelling/No pain) and 100 (Worst possible swelling/Worst possible pain). The AMRA-5 is a composite score of the 5 major HAE symptoms: skin swelling, skin pain, abdominal pain, voice change and difficulty swallowing. For non-severe laryngeal attacks, the AMRA-5 scale allowed scoring on a scale between 0 (No skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing) and 100 (Worst skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing). Time to substantial symptom relief is calculated as the time from study drug administration for treating an attack (the time of the first dose if a second dose is administered) to the earliest time when a ≥50% reduction from pre-treatment in AMRA is achieved and a composite score from pre-treatment for 2 consecutive timepoints within 12 hours post-treatment. If the event of interest has not been achieved based on data entered within the timeframe, an attack will be treated as right-censored at the last observation before 13h, which is the upper end of the ePRO data entry window for 12h timepoint. The Kaplan-Meier estimated median and percentiles of time to substantial symptom relief by Angioedema Symptom Rating Scale assessment has been presented.

	The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 12 hours post-treatment

Analysis Population Description: PEAS.

End point values		**Placebo**	**Deucrictibant IR 20 mg**
Number of subjects analysed		88	88
Units: Hours			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	NE (NE, NE)	3.37 (2.56, 4.95)

NE= Not Estimable indicates that median was not estimable as less than 50% achieved the endpoint within 12-hour analysis window

9. Secondary Endpoint

Endpoint Title	Time to almost complete or complete symptom relief by AMRA assessment
Endpoint Description	AMRA-3 is a composite score (average) of the 3 major HAE symptoms: skin swelling, skin pain, and abdominal pain. The AMRA-3 scale allowed scoring on a scale between 0 (No swelling/No pain) and 100 (Worst possible swelling/Worst possible pain). The AMRA-5 is a composite score of the 5 major HAE symptoms: skin swelling, skin pain, abdominal pain, voice change and difficulty swallowing. For non-severe laryngeal attacks, the AMRA-5 scale allowed scoring on a scale

	between 0 (No skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing) and 100 (Worst skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing). Time to almost complete or complete symptom relief is calculated as the time from study drug administration for treating an attack (the time of the first dose if a second dose is administered) to the earliest time when all individual items of the AMRA assessment achieve values ≤10 and is sustained for two consecutive timepoints. If the event of interest has not been achieved based on data entered within the timeframe, an attack will be treated as right-censored at the last observation before 30h, which is the upper end of the ePRO data entry window for 24h timepoint. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 24 hours post-treatment

Analysis Population Description: PEAS.

End point values		**Placebo**	**Deucrictibant IR 20 mg**
Number of subjects analysed		88	88
Units: Hours			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	NE (NE, NE)	6.94 (4.70, 13.44)

NE= Not Estimable indicates that median was not estimable as less than 50% achieved the endpoint within 24-hour analysis window

10. Secondary Endpoint

Endpoint Title	Percentage of participants with study drug treated attacks who reached almost complete or complete symptom relief by AMRA assessment
Endpoint Description	The AMRA-3 is a composite score (average) of the 3 major HAE symptoms: skin swelling, skin pain, and abdominal pain. The AMRA-3 scale allowed scoring on a scale between 0 (No swelling/No pain) and 100 (Worst possible swelling/Worst possible pain). The AMRA-5 is a composite score of the 5 major HAE symptoms: skin swelling, skin pain, abdominal pain, voice change and difficulty swallowing. For non-severe laryngeal attacks, the AMRA-5 scale allowed scoring on a scale between 0 (No skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing) and 100 (Worst skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing). A study drug treated attack, if the last or the second-to-last available AMRA assessment prior to 24h has all individual item scores ≤ 10 and is sustained for two consecutive timepoints, is considered achieved almost complete or complete symptom relief by AMRA at 24 hours post-treatment. This binary endpoint ("Yes" or "No") was assigned a value of "Yes" for a study drug–treated attack if the last or the second-to-last available AMRA assessment prior to 24 hours had all individual item scores ≤10 and this status was sustained for two consecutive time points. In such cases, the outcome of almost complete or complete symptom relief by AMRA was considered to have been achieved at 24 hours post-treatment. The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 24 hours post-treatment

Analysis Population Description: PEAS.

End point values	Placebo	Deucrictibant IR 20 mg
Number of subjects analysed	88	88

Summary of Results

End point values		Placebo	Deucrictibant IR 20 mg
Units: Percentage of participants			
Measure Type	**Precision/ Dispersion type**		
Number	Not Applicable	34.1	75.0

11. Secondary Endpoint

Endpoint Title	Time to EoP in attack symptoms by AMRA assessment
Endpoint Description	AMRA-3 is a composite score (average) of the 3 major HAE symptoms: skin swelling, skin pain, and abdominal pain. The AMRA-3 scale allowed scoring on a scale between 0 (No swelling/No pain) and 100 (Worst possible swelling/Worst possible pain). The AMRA-5 is a composite score of the 5 major HAE symptoms: skin swelling, skin pain, abdominal pain, voice change and difficulty swallowing. For non-severe laryngeal attacks, the AMRA-5 scale allowed scoring on a scale between 0 (No skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing) and 100 (Worst skin swelling, skin pain, abdominal pain, voice change, or difficulty swallowing). Time to EoP in attack symptoms is calculated as the time from study drug administration for treating an attack (the time of the first dose if a second dose is administered) to the earliest post-treatment timepoint after which all subsequent AMRA results are stable or improved, namely, at all subsequent timepoints, for the AMRA assessment, each individual item score value is equal to or lower (improved) than the corresponding item score value at the current time (the identified time of EoP). The ICE was the intake of rescue medication or prohibited medication after the attack was treated with study drug. The occurrence of ICE was regarded as a treatment failure.
Time Frame	Up to 12 hours

Analysis Population Description: PEAS.

End point values		Placebo	Deucrictibant IR 20 mg
Number of subjects analysed		88	88
Units: Minutes			
Measure Type	**Precision/ Dispersion type**		
Median	95% Confidence Interval	272.09 (147.57, NE)	25.25 (17.82, 30.25)

NE= Not Estimable indicates that median was not estimable as less than 50% achieved EoP in attack symptoms within 12-hour analysis window

E. ADVERSE EVENTS

1. ADVERSE EVENTS INFORMATION

Timeframe for reporting adverse events: Up to 18 months

Adverse event reporting additional description: Treatment emergent non serious adverse events (TENSAEs) and Treatment emergent serious adverse events (TESAEs) were collected in Safety Analysis Set which comprises of all enrolled participants who received any dose of study drug.

Assessment type	non-systematic

Dictionary used

Dictionary name	MedDRA
Dictionary version	28.0

2. REPORTING GROUPS

Reporting group title	Reporting group description
Placebo	Participants received matching placebo orally after an HAE attack in a crossover manner. Safety analysis was performed by actual treatment received.
Deucrictibant IR 20 mg	Participants received deucrictibant IR 20 mg soft capsule orally, administered in a crossover manner following an HAE attack. Additionally, participants ≥12 to <18 years of age received deucrictibant IR 20 mg soft capsule orally in the absence of an attack for assessment of PK and safety.

3. ALL-CAUSE MORTALITY

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
	Affected/At Risk (%)	Affected/At Risk (%)	Affected/At Risk (%)
Total All-Cause Mortality	0/101 (0 %)	0/100 (0 %)	0/10 (0 %)

4. SERIOUS ADVERSE EVENTS

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Total subjects affected by serious adverse events **Subjects affected/exposed**	1/101 (1.0%)	1/100 (1.0%)	0/10 (0%)
Number of deaths (all causes)	0	0	0
Number of deaths resulting from adverse events	0	0	0
Congenital, familial and genetic disorders			
Hereditary angioedema			
Subjects affected / exposed	0/101(0%)	1/100 (1.0%)	0/10 (0%)
Occurrences causally related to treatment / all	0/0	0/1	0/0
Deaths causally related to treatment / all	0/0	0/1	0/0
Pregnancy, puerperium and perinatal conditions			
Abortion spontaneous			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Total subjects affected by serious adverse events **Subjects affected/exposed**	1/101 (1.0%)	1/100 (1.0%)	0/10 (0%)
Number of deaths (all causes)	0	0	0
Number of deaths resulting from adverse events	0	0	0
Occurrences causally related to treatment / all	0/1	0/0	0/0
Deaths causally related to treatment / all	0/0	0/0	0/0

5. NON-SERIOUS ADVERSE EVENTS

Frequency threshold for reporting non-serious adverse events: 0%

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Total	17/101 (16.8%)	24/100 (24.0%)	3/10 (30.0%)
Cardiac disorders			

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Atrial fibrillation			
Subjects affected / exposed	1/101(1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Ear and labyrinth disorders			
Ear pain			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Eye disorders			
Vision blurred			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Gastrointestinal disorders			
Abdominal pain			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	1/10 (10.0%)
Occurrences (all)	0	1	1
Abdominal pain upper			

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Subjects affected / exposed	1/101 (1.0)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	1	1	0
Diarrhoea			
Subjects affected / exposed	0/101 (0%)	2/100 (2.0%)	0/10 (0%)
Occurrences (all)	0	2	0
Dyspepsia			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Flatulence			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Vomiting			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
General disorders and administration site conditions			
Fatigue			

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Subjects affected / exposed	0/101 (0%)	3/100 (3.0%)	0/10 (0%)
Occurrences (all)	0	3	0
Influenza like illness			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Lethargy			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Hepatobiliary disorders			
Cholelithiasis			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Infections and infestations			
Ear infection			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Gastroenteritis			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Gastrointestinal infection			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Influenza			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Nasopharyngitis			
Subjects affected / exposed	1/101 (1.0%)	2/100 (2.0%)	0/10 (0%)
Occurrences (all)	1	2	0
Respiratory tract infection			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Tonsillitis			

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Upper respiratory tract infection			
Subjects affected / exposed	2/101 (2.0%)	1/100 (1.0%)	1/10 (10.0)
Occurrences (all)	2	1	1
Injury, poisoning and procedural complications			
Chest injury			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Contusion			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Fall			
Subjects affected / exposed	0/101 (0%)	2/100 (2.0%)	0/10 (0%)
Occurrences (all)	0	2	0

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Ligament injury			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Ligament sprain			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Skin wound			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Investigations			
Blood cholesterol increased			
Subjects affected / exposed	2/101 (2.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	2	0	0
International normalised ratio increased			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Occurrences (all)	1	0	0
Musculoskeletal and connective tissue disorders			
Back pain			
Subjects affected / exposed	0/101 (0%)	3/100 (3.0%)	0/10 (0%)
Occurrences (all)	0	3	0
Myalgia			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Sacral pain			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Neoplasms benign, malignant and unspecified (including cysts and polyps)			
Bladder transitional cell carcinoma			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Endometrial cancer			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Nervous system disorders			
Brain fog			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Headache			
Subjects affected / exposed	0/101 (0%)	2/100 (2.0%)	1/10 (10.0%)
Occurrences (all)	0	2	1
Somnolence			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Syncope			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Reproductive system and breast disorders			
Dysmenorrhoea			
Subjects affected / exposed	0/101 (0%)	2/100 (2.0%)	1/10 (10.0%)
Occurrences (all)	0	2	1
Endometrial hyperplasia			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Respiratory, thoracic and mediastinal disorders			
Asthma			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Cough			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Skin and subcutaneous tissue disorders			
Dermatitis			

Summary of Results

	Study drug treated attacks		Non-attack
	Placebo	**Deucrictibant IR 20 mg**	**Deucrictibant IR 20 mg**
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Pruritus			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0
Rash			
Subjects affected / exposed	0/101 (0%)	1/100 (1.0%)	0/10 (0%)
Occurrences (all)	0	1	0
Vascular disorders			
Hypertension			
Subjects affected / exposed	1/101 (1.0%)	0/100 (0%)	0/10 (0%)
Occurrences (all)	1	0	0

F. ADDITIONAL INFORMATION

1. GLOBAL SUBSTANTIAL MODIFICATIONS

Were there any global substantial amendments to the protocol? Yes

Amendment Date	Amendment Description
Amendment 1.0, Date: 01 Feb 2024 (Global)	• The ranking of the secondary efficacy endpoints was revised, such that endpoints based on PGI-C and PGI-S were ranked above those based on AMRA. • The secondary endpoint of time to EoP in attack symptoms within 12 hours was clarified to specify that, with EoP, time was defined as the earliest post-treatment timepoint after which all subsequent PGI-C ratings are stable or improved. • In Inclusion Criterion 3, the minimum age for onset of HAE symptoms, if applicable as a criterion for confirmation of diagnosis, was to occur before age ≤30 years rather than 40 years. • In Inclusion Criterion 8 and other applicable sections of the protocol, it was clarified that the body weight requirement of ≥40 kg was applicable to adolescent participants (ie, those ≥12 to <18 years of age). • Exclusion criterion 6 was added to exclude participants with abnormal renal function (estimated glomerular filtration rate CKD-EPI <60 mL/min/1.73 m2). • It was indicated that the dates and times of concomitant medication administration were to be documented. It was clarified that the End-of-Study Visit included both on-site and remote assessments.
Amendment 2.0, Date: 19 May 2024 (Global)	• The study population was revised to include approximately 30% of participants who are using long-term prophylactic medication for HAE globally; in the United States (including Puerto Rico), enrollment is targeted to be approximately 50% of participants using long-term prophylactic medication. The inclusion criteria were revised accordingly.

	• After the on-site Screening Visit, it was clarified that the Investigator will confirm whether the participant is eligible for the study after the results of all safety laboratory tests have become available. • If a participant opts to have their Day 1 visit performed remotely, it was clarified randomization should take place one day before the Day 1 visit, so that study drug can be shipped to the participant's home in time for the Day 1 visit. • Inclusion criterion 3 was modified to indicate that if treatment washout is required, the confirmatory C1-INH testing should be performed at least 5 half- lives after the last dose of C1-INH or attenuated androgens therapy. • Inclusion criterion 9 was clarified to indicate that female participants of child-bearing potential are to use acceptable methods of birth control as defined in the protocol and as available locally. • It was clarified that use of concomitant medications with systemic absorption that are strong inhibitors/inducers of CYP3A4, such as clarithromycin, itraconazole, ketoconazole, and ritonavir within the last 30 days, or within 5 half-lives (whichever is longer) of the time of randomization are a criterion for exclusion and are prohibited during the study. • The window for the AMRA interview was increased from 48 to 72 hours after the Day 1 visit.
Amendment 1.1, Date: 21 Jun 2024 (EU only)	• It was added that a designee besides the Investigator could qualify an attack and that a worksheet would be provided. • Text was added to the Quality Control and Quality Assurance section to state the reporting requirements for serious breaches. Serious breaches were to be reported in

	accordance with EU/European Economic Area Regulation No 536/2014, Article 52. It was clarified that the End-of-Study Visit included both on-site and remote assessments.
Amendment 1.2, 03 July 2024 (EU only)	• It was specified that highly effective (rather than medically acceptable) contraception was to be used by female participants of childbearing potential during the study. It was also clarified that the contraception methods used were to be based on the approved methods in the participant's country. • Text was added to state that more frequent pregnancy testing could be performed if required by local regulations or in the case of a suspected pregnancy. • The term tubal ligation was changed to bilateral salpingectomy.
Amendment 3.0, 28 October 2024 (Global)	• It was noted that adolescent participants (≥12 to <18 years of age) were to be enrolled in accordance with local age requirements and details regarding attainment of written informed consent from adolescent participants were clarified. Information on blood volume requirements was added. • Exclusion criterion regarding use of attenuated androgens was modified. • The procedure for clinical laboratory assessment, including safety and pregnancy testing, was clarified to allow for more frequent pregnancy testing as per local requirements and also allow for laboratory tests to be assessed locally in parallel with the central laboratory. • The contraceptive language was revised globally to align with previously implemented local amendments. • Benefit-risk language was augmented. • The Safety Analysis Set was revised to include all enrolled (rather than randomized)

	participants who received any dose of study drug.
Amendment 4.0, 15 January 2025 (Global)	• The primary reason for the amendment was to allow enrollment of participants with HAE type 3 (HAE-nC1INH).
Amendment 4.1, 17 January 2025 (EU only)	• The primary reason for this amendment was to align with the previously implemented global amendment. Changes made to this amendment included changes made in prior global protocol amendments: Amendment 3.0 (dated 28 October 2024) and Amendment 2.0 (dated 19 May 2024). • Text was updated to allow enrollment of participants with HAE type 3 (HAE-nC1INH). • Study drug treatment of qualifying non-severe laryngeal attacks was to be permitted in adult participants only (note that qualifying non-severe laryngeal attacks were allowed to be treated in adolescent participants outside of the EU).

2. GLOBAL INTERRUPTIONS AND RE-STARTS

Were there any global interruptions to the trial? No

3. LIMITATIONS AND CAVEATS

None

4. PUBLICATIONS

5. A DECLARATION BY THE SUBMITTING PARTY ON THE ACCURACY OF THE SUBMITTED INFORMATION

The Sponsor certifies that the information provided in this report is accurate to the best of their knowledge and that they have performed due diligence in its collection and reporting. The sponsor further certifies that there are no exaggerations, falsifications, misrepresentations, omissions, misleading information, or negative findings.

The Sponsor understands that this report will be subject to verification, and that any exaggeration, falsification, misrepresentation, omission, provision of misleading information, or any other negative finding will be cause for reprimand under European Union law.